COMP S492F/3920SEF
Machine Learning

Unsupervised learning

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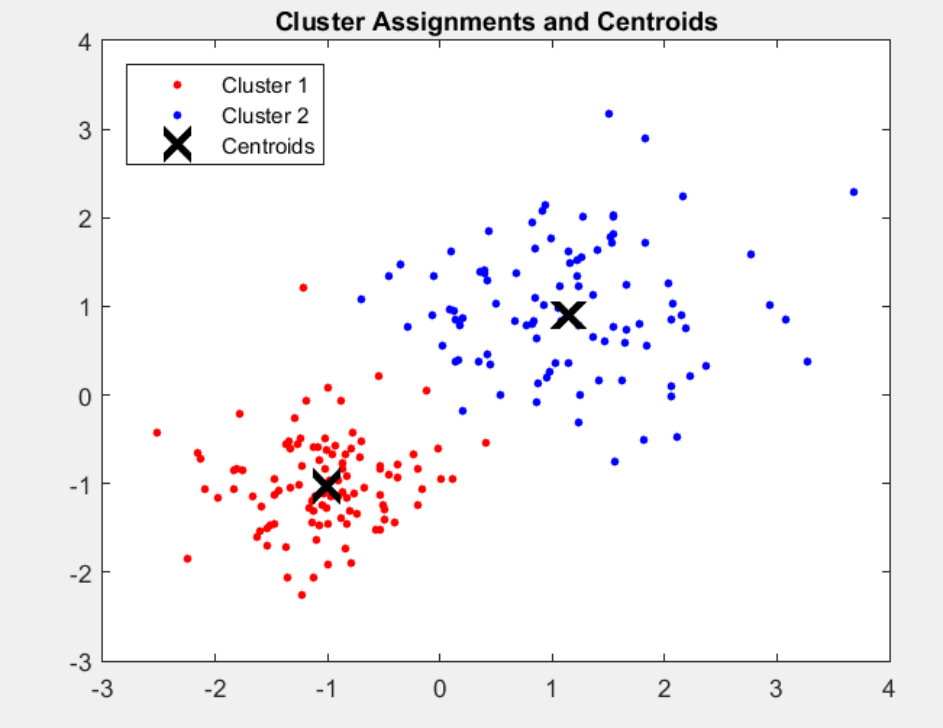
Contents

- K-Means Clustering
- Basic Techniques and Related Issues in Machine Learning

K-Means Clustering

- Non-parametric
- Unsupervised learning algorithm
- Groups the given objects into K number of groups with similar attributes/features



A cluster is a collection of data points aggregated together due to certain similarities.

A centroid is the imaginary or real location representing the center of the cluster.

Environment Perception in Automatic Vehicle

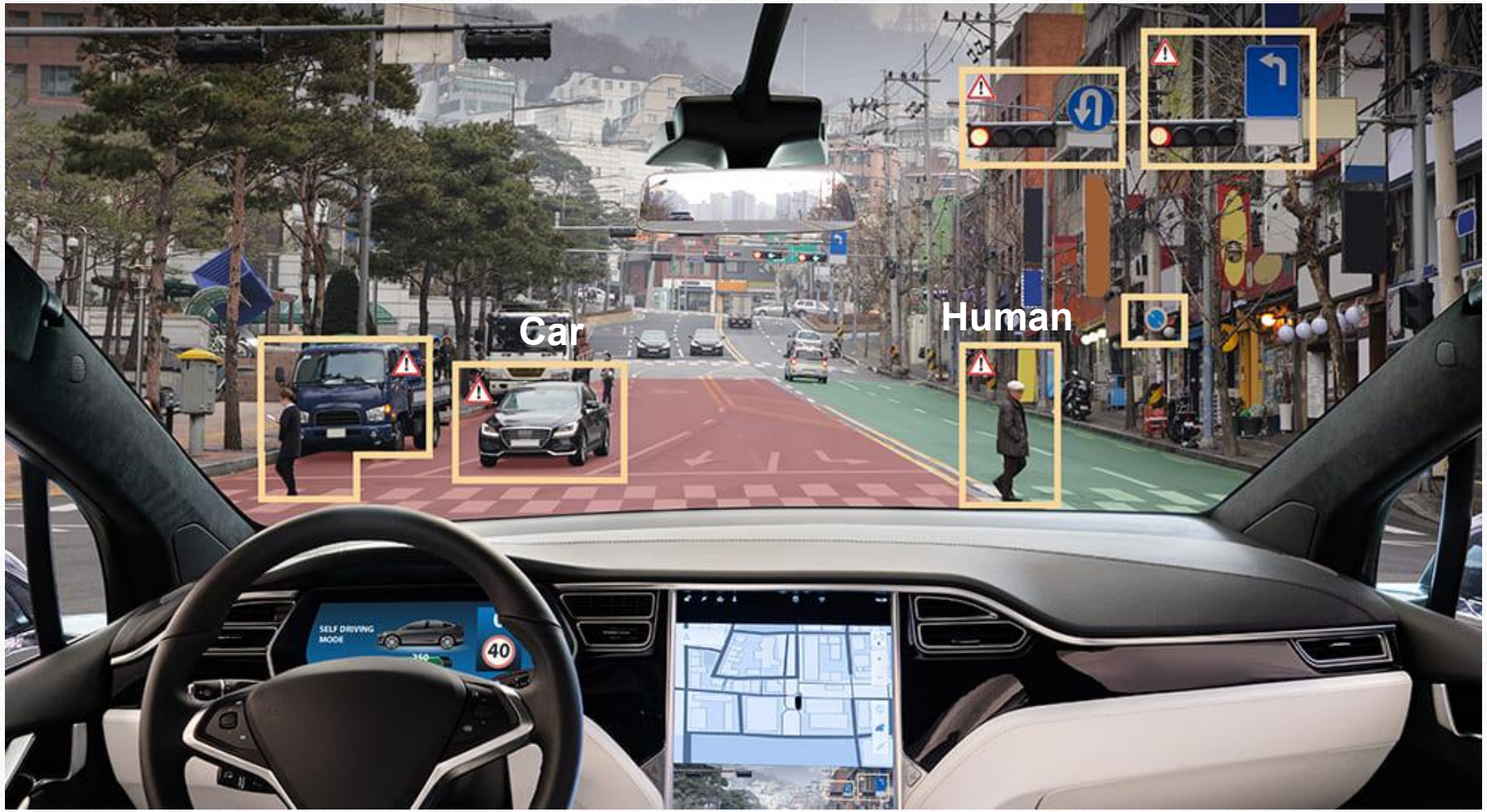


Image Segmentation

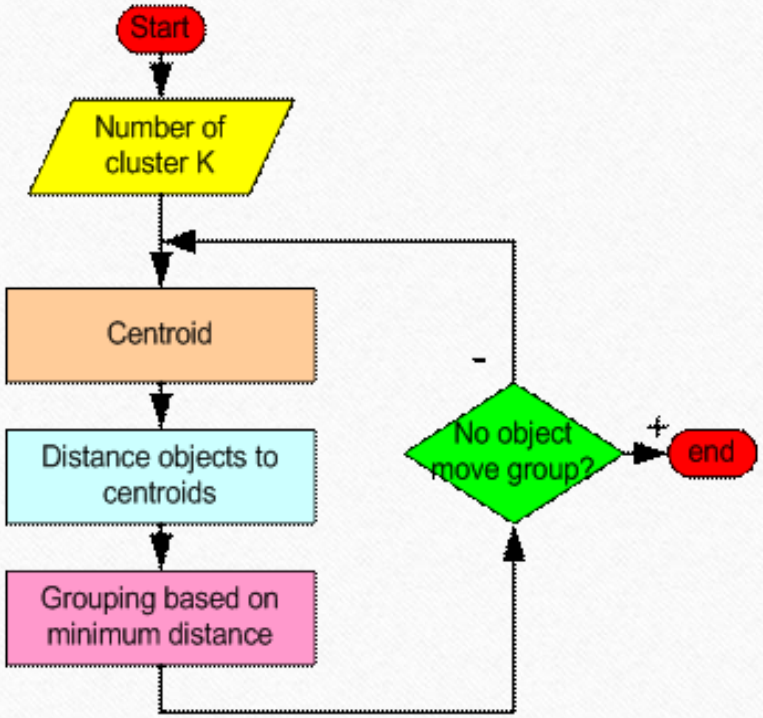
K-Means Clustering

Initialization

- Choose a value of K.
- Choose K random positions in the input space.
- Assign the cluster centroids to those positions.

Learning

- Repeat
  - For each data point
    - Compute the distance to each cluster centroid.
    - Assign the data point to the nearest centroid.
  - For each cluster centroid
    - Move the position of the centroid to the distance mean of the data points in the cluster.
- Until no objects move to other groups and the cluster centroids stop moving.



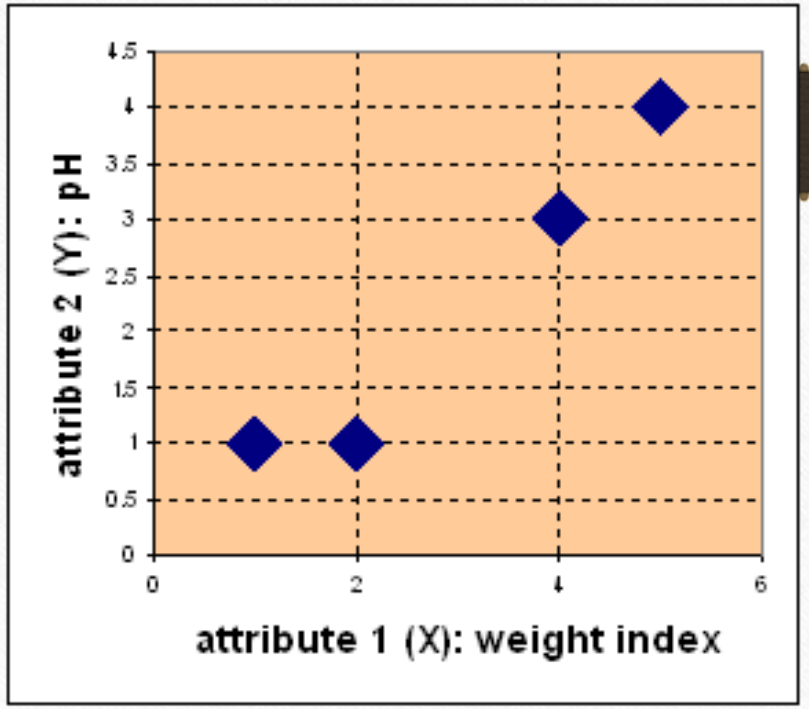
Worked Example

Problem: 4 drugs A, B, C and D and each drug has 2 attributes:

Goal: classify these drugs into 2 different groups (i.e. K = 2).

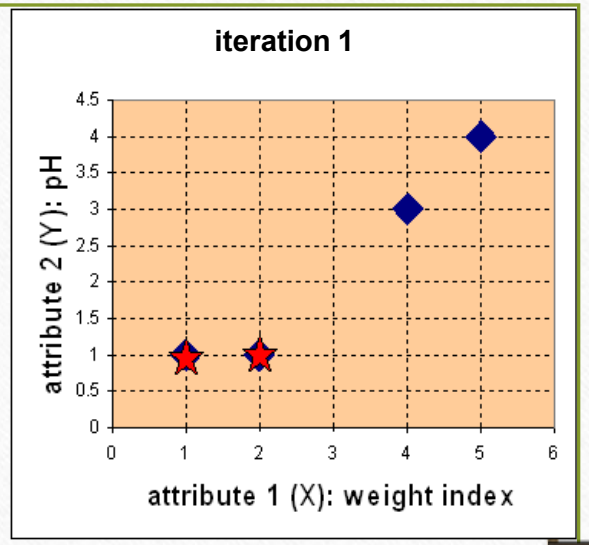
Object	Attribute X (weight index)	Attribute Y (pH)
Drug A	1	1
Drug B	2	1
Drug C	4	3
Drug D	5	4

The 4 drugs can be represented by 4 coordinate points based on their attributes:



Worked Example (1st iteration)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \begin{matrix} X \\ Y \end{matrix}$$



Step 1.1: **Choose the centroids**

- Drug A and drug B are chosen as the first centroids with coordinates c1=(1, 1) and c2=(2, 1) respectively.

Step 1.2: **Find the distances of objects from centroids**

- Then we calculate the Euclidean distances of all objects to each centroid and the results can be represented by a distance matrix D1:

$$\mathrm{D1} = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 1 & 0 & 2.83 & 4.24 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) \text{ group } 1 \\ c_2 = (2,1) \text{ group } 2 \end{matrix}$$

A B C D

- Each column symbolizes an object. The 1st row corresponds to the distance of each object to the 1st centroid while the 2nd row gives the distance of each object to the 2nd centroid.

Step 1.3: **Group the objects**

- Then we assign each object to a group based on the minimum distance. The result is given by a group matrix G1:

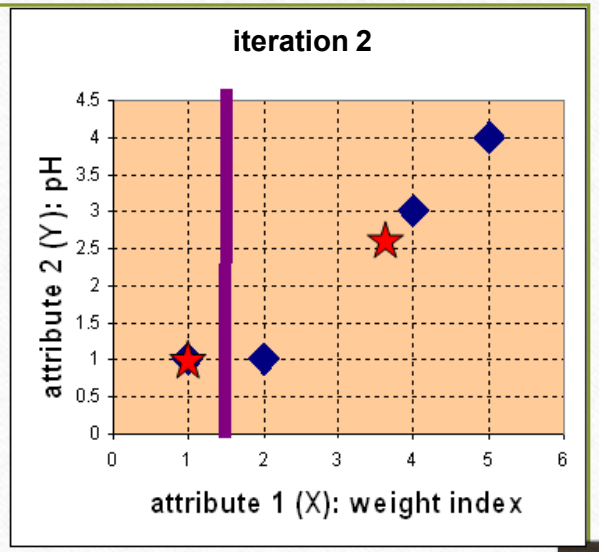
$$\mathrm{G1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) \text{ group } 1 \\ c_2 = (2,1) \text{ group } 2 \end{matrix}$$

A B C D

- Note that the element of the group matrix is 1 if the object is assigned to that group and is 0 if not assigned.

Worked Example (2nd iteration)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \begin{matrix} X \\ Y \end{matrix}$$



Step 2.1: **Update the positions of centroids**

- Next we have to compute the new centroid of each group based on these new memberships. Group 1 has only one member thus it remains at c1=(1, 1). Group 2 has 3 members and the new centroid is just the average coordinates of its 3 members:

$$c_2 = \left(\frac{2+4+5}{3}, \frac{1+3+4}{3}\right) = (\frac{11}{3}, \frac{8}{3})$$

Step 2.2: **Find the distances of objects from centroids**

- The next step is to compute the distance of all objects to the new centroids. The new distance matrix D2 is found:

$$\mathrm{D2} = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 3.14 & 2.36 & 0.47 & 1.89 \end{bmatrix} \begin{matrix} c_1 = (1,1) & \text{group 1} \\ c_2 = (11/3, 8/3) & \text{group 2} \end{matrix}$$

A B C D

Step 2.3: **Regroup the objects**

- Similarly, we assign each object to a group based on the minimum distance. The result is given by a group matrix G2:

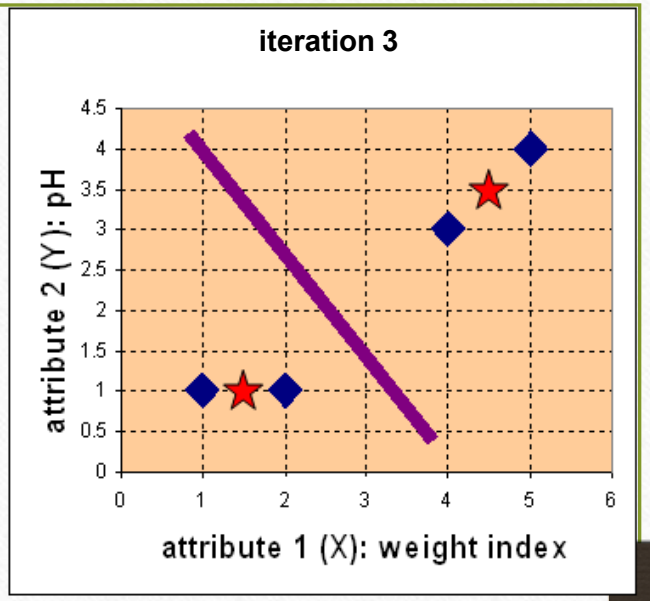
$$\mathrm{G2} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \text{group 1} \\ \text{group 2} \end{matrix}$$

A B C D

- Note that the drug B has moved to group 1 while the other drugs remain.

Worked Example (3rd iteration)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \begin{matrix} X \\ Y \end{matrix}$$



Step 3.1: **Update the positions of centroids**

- Now we have to compute the coordinates of the new centroids.

$$c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1\frac{1}{2}, 1) \qquad c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4\frac{1}{2}, 3\frac{1}{2})$$

Step 3.2: **Find the distances of objects from centroids**

- Repeating step 2, we get the new distance matrix D3.

$$\mathrm{D3} = \begin{bmatrix} 0.5 & 0.5 & 3.20 & 4.61 \\ 4.30 & 3.54 & 0.71 & 0.71 \end{bmatrix} \quad \begin{matrix} c_1 = (1.5,1) & \text{group 1} \\ c_2 = (4.5, 3.5) & \text{group 2} \end{matrix}$$

A B C D

Step 3.3: **Regroup the objects**

- We assign each drug base on their minimum distances from the new centroids.
- We get the result: G3 = G2, meaning that no more drugs move to other groups and the clustering has reached its stability. The final grouping table:

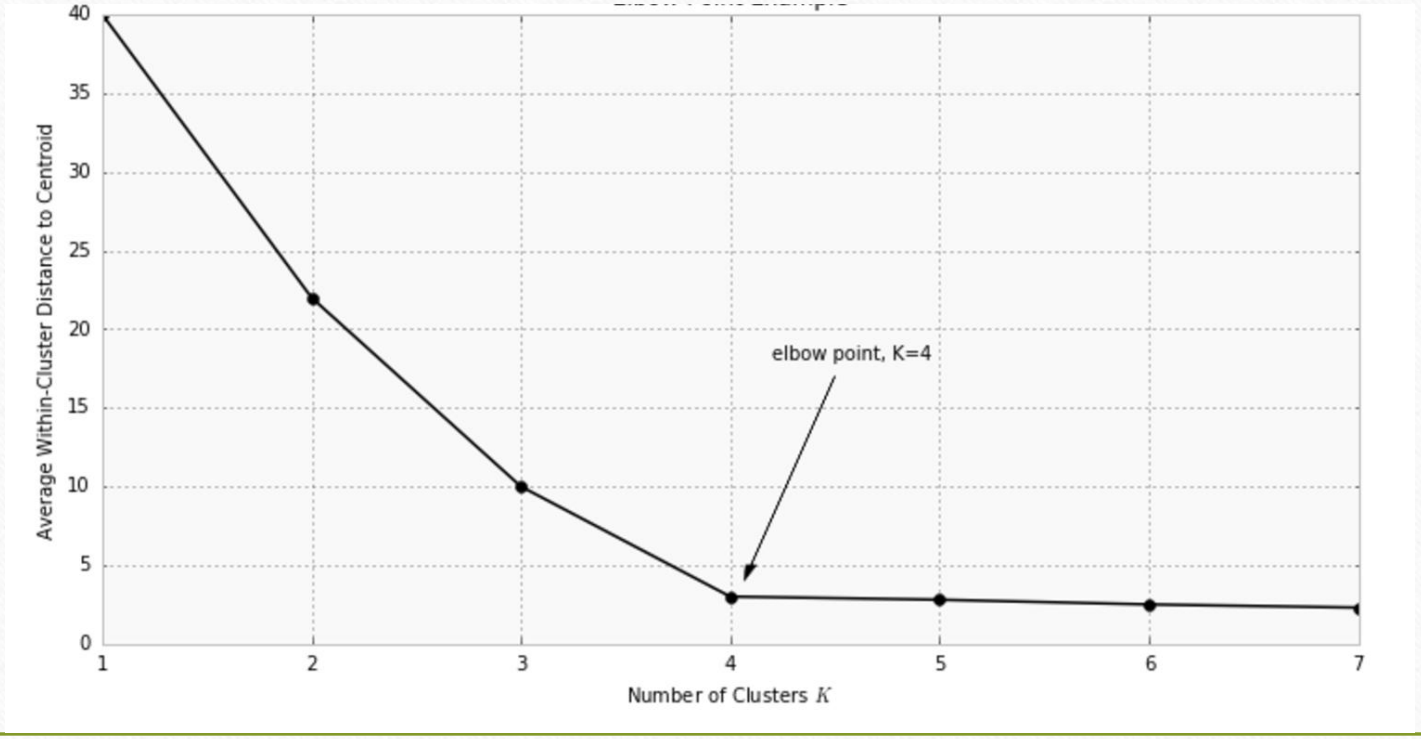
$$\mathrm{G3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{matrix} \text{group 1} \\ \text{group 2} \end{matrix}$$

A B C D

Object	Attribute X (weight index)	Attribute Y (pH)	Group assigned
Drug A	1	1	1
Drug B	2	1	1
Drug C	4	3	2
Drug D	5	4	2

Choice of K

- In general, there is no method for determining the exact value of K.
- Compare results across different values of K: measure the *average of mean distance between data points and their cluster centroid of all clusters* (i.e. a measure of compactness of the clusters).
- The average mean distance to the centroid of clusters as a function of *K* is plotted and the "*elbow point*“, where the rate of decrease sharply shifts, can be used to roughly determine *K*.



Strength and Weakness

- **Strength:**
  - Work well with *globular* clusters (i.e. each cluster has a roughly spherical or elliptical shape in the input space) and in large data collection.
  - Simple and easy to implement.
  - Fast and efficient in terms of computational cost.
- **Weakness:**
  - Different initial partitions can result in different final clusters. https://www.youtube.com/watch?v=9nKfViAfajY
  - It takes time to run the program in advance to obtain an optimal K by using different K values for comparison.
  - May work poorly with clusters which have very different sizes and different densities.
  - Often produce clusters with relatively uniform size even if the input data have different cluster size.
  - The assumption that each attribute has the same contribution might not be satisfied.
  - Sensitive to outliers.

Outliers are considered as observations that should be removed in order to make clustering more reliable .

Practical Applications

- **E-commerce**: Cluster products based on customers' ratings in order to optimize the purchase-profit ratio of the enterprise in e-commerce websites.
- **Grouping documents**: Form groups of documents belonging to same/similar topic.
- **Grouping images**: e.g. Google image search.
- **Data compression**: Colors present in the image can be quantized and will further help in compressing the image.

Basic Techniques and Related Issues in Machine Learning

Detailed ML Workflow

A proper machine learning workflow includes:

- Data collection
- Data preprocessing
- Separate training and test sets
- Try appropriate algorithms
- Fit model parameters (for parametric methods)
- Tune the hyperparameters
- Model evaluation by selecting a proper performance metric
- A systematic cross-validation
- Attain an optimal balance of bias and variance which leads to a model that is neither overfit nor underfit.

In different stages, some basic techniques are used, e.g. data preprocessing (feature scaling, selection and extraction), model evaluation (using precision and recall for classifiers) and K-fold cross validation.

Feature Selection

Reasons to use feature selection

- It enables the machine learning algorithm to train faster.
- It reduces the complexity of a model and makes it easier to interpret.
- It improves the accuracy of a model if the optimal subset is chosen.
- It reduces overfitting.

Via regularization (implicit)

- Regularization methods such as LASSO (L1) regression and RIDGE (L2) regression have inherent penalization functions to reduce overfitting as we have seen before.

Univariate feature selection (explicit)

- It examines each feature individually to determine its relation to the response variable.
- One of the simplest method used is **Pearson correlation coefficient**, which measures the linear correlation between two variables. The coefficient lies between -1 to 1:
  - -1: perfect negative correlation (one variable increases as the other decreases)
  - 0: no linear correlation between the two variables
  - 1: perfect positive correlation (one variable increases as the other increases)

Pearson Correlation Coefficient

Find the value of the *Pearson correlation coefficient* from the table showing a dataset consisting of 6 persons with their ages and glucose levels:

SUBJECT	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

Pearson Correlation Coefficient: Example

- Step 1:

SUBJECT	AGE X	GLUCOSE LEVEL Y	XY	X^2	Y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

- Step 2: **Pearson correlation coefficient** $= \dfrac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,]\,[\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$

Pearson correlation coefficient = [6(20,485) – (247 × 486)] / [√[[6(11,409) – 247^2] × [6(40,022) – 486^2]]] = 0.5298

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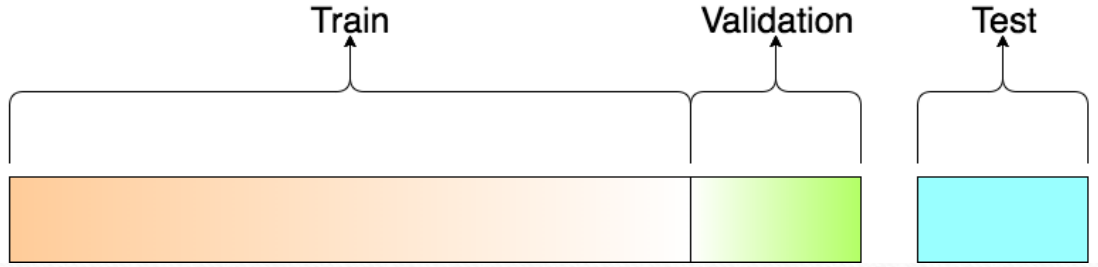
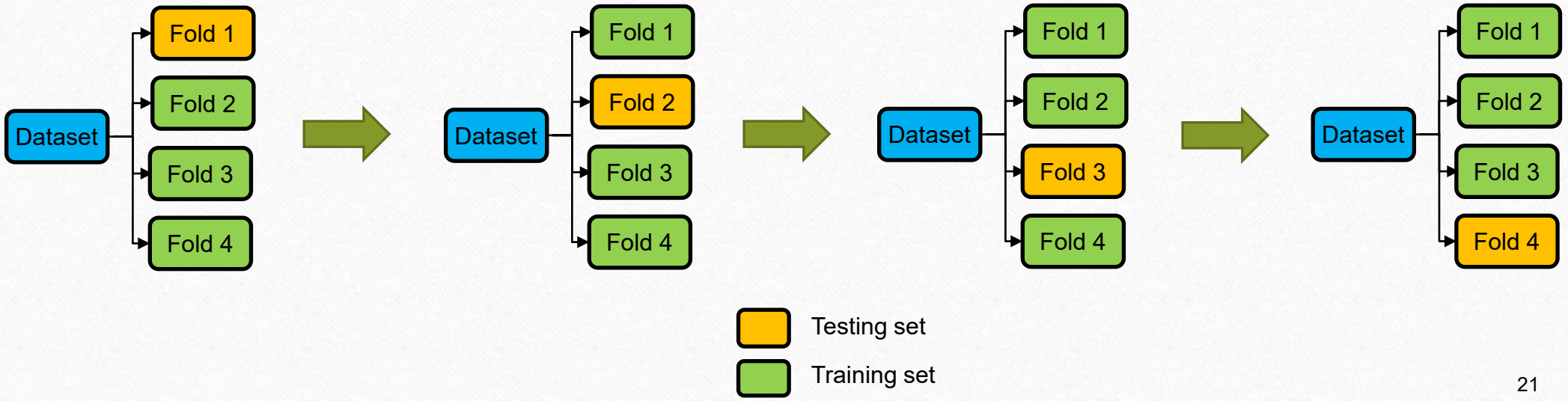
Training, Validation and Test Sets

- The data used to build the final model usually comes from multiple datasets. In particular, three datasets are commonly used in different stages of the creation of the model.
- The model is initially fit on a **training dataset**, that is a set of examples used to fit the parameters (e.g. weight coefficients) of the model.
- Successively, the fitted model is used to predict the responses for the observations in a second dataset called the **validation dataset** whose data is unseen before. Hyperparameter tuning can be done in this stage.
- Finally, the **test dataset** is a dataset used to provide an unbiased evaluation of a final model fit on the training dataset.
- **Dataset split ratio**: It depends on the total number of samples in the data and also on the actual model for training. One common practice: training and test (80/20) and then the training set is further split into training (actual) and validation (again, 80/20).



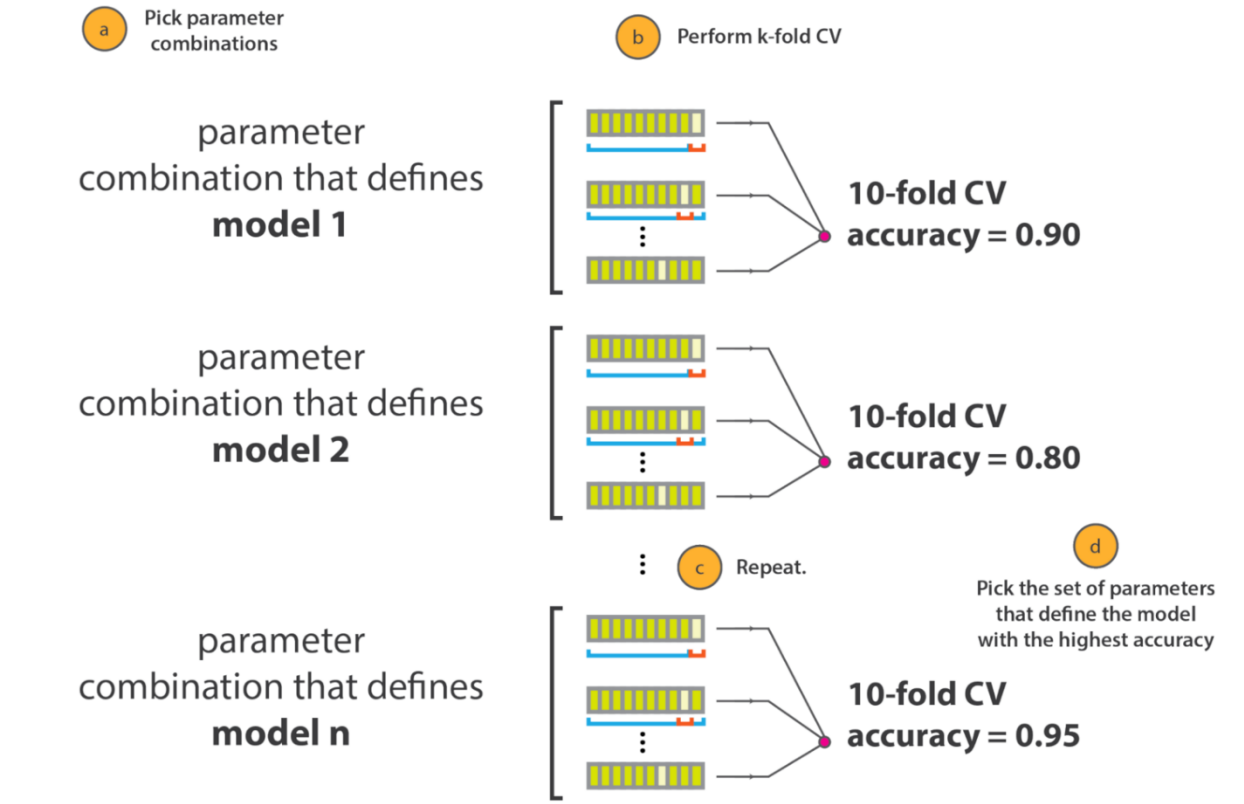
K-Fold Cross Validation

- *Cross-validation* is a technique used to protect against overfitting in a predictive model, particularly in case the amount of data may be limited.
- In **K-fold cross validation**, a fixed number, *K*, of folds (or partitions) of the training set is made. One fold is taken as the validation/test set in turn. The analysis is run on the remaining *K* – 1 folds (the train set) for *K* times and the average error estimate is computed.



The Use of K-Fold Cross Validation

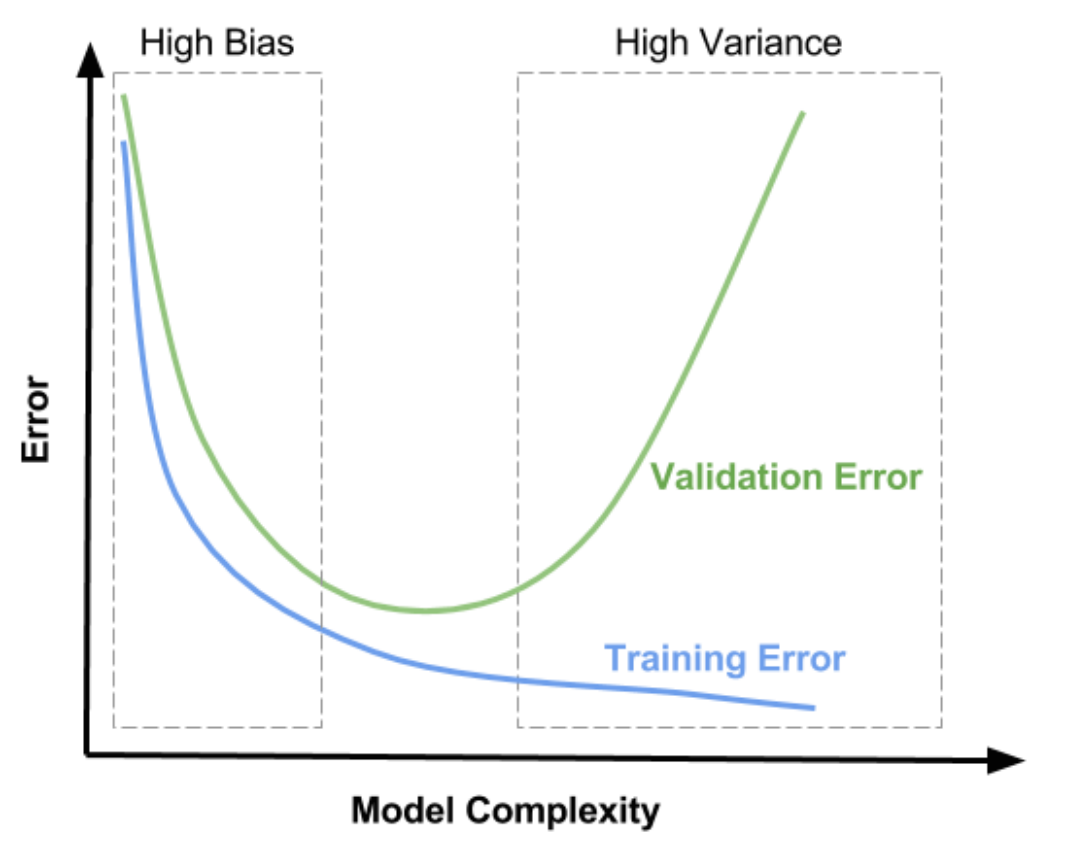
- **Hyperparameter tuning**: An illustration of using the 10-fold cross validation for *n* models with different set of hyperparameters:



- Similarly, *K-fold cross validation* can be used to evaluate models with different algorithms (e.g. logistic regression and decision tree) for model selection.

The Use of K-Fold Cross Validation

- *K-fold cross validation* can be used for *regularization* by *early stopping*: stop training when the error on the validation dataset increases, as this is a sign of overfitting to the training dataset.



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Types of Prediction Errors

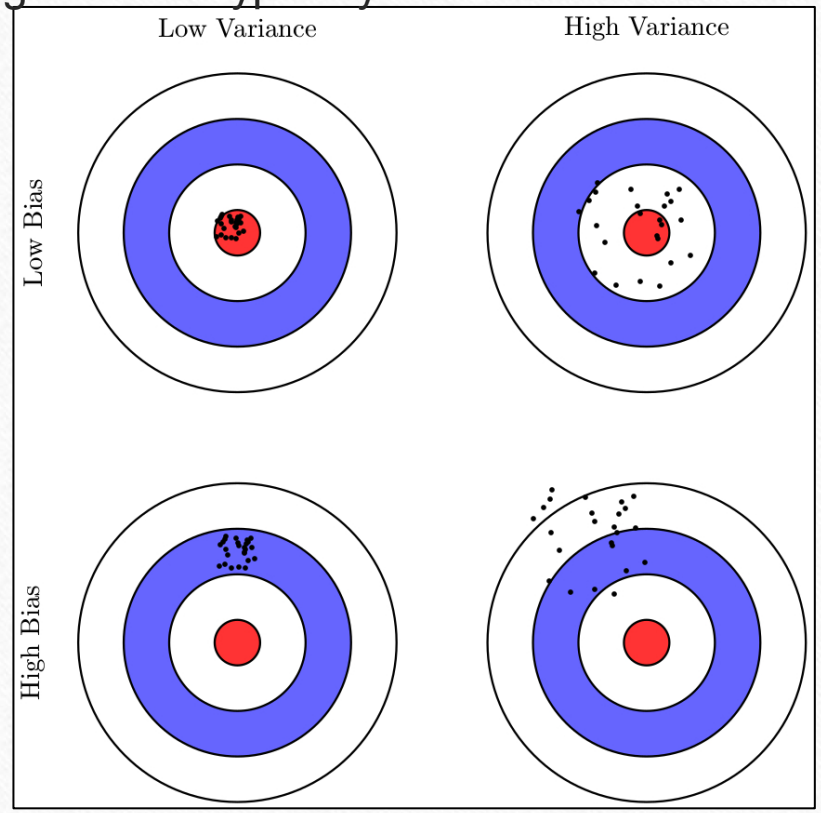
Prediction error measures how well the model predicts the response variable **in regression**;

or samples are classified to the correct category **in classification**.

1. **Irreducible error (noise)**: It can't be reduced by the choice in algorithm. It typically comes from inherent randomness, or an incomplete feature set.

The other two types of errors, however, can be reduced because they stem from the choice of algorithm used for predictions.

2. **Bias:** It is an error from erroneous assumptions in the learning algorithm. Some algorithms are simply too rigid to learn complex signals from the dataset, e.g. fitting a linear regression to a dataset that has a non-linear pattern. It causes *underfitting*.
3. **Variance:** It is an error from sensitivity to small fluctuations in the training set. High variance can cause an algorithm to model the random noise in the training data, rather than the intended outputs. It causes *overfitting*.

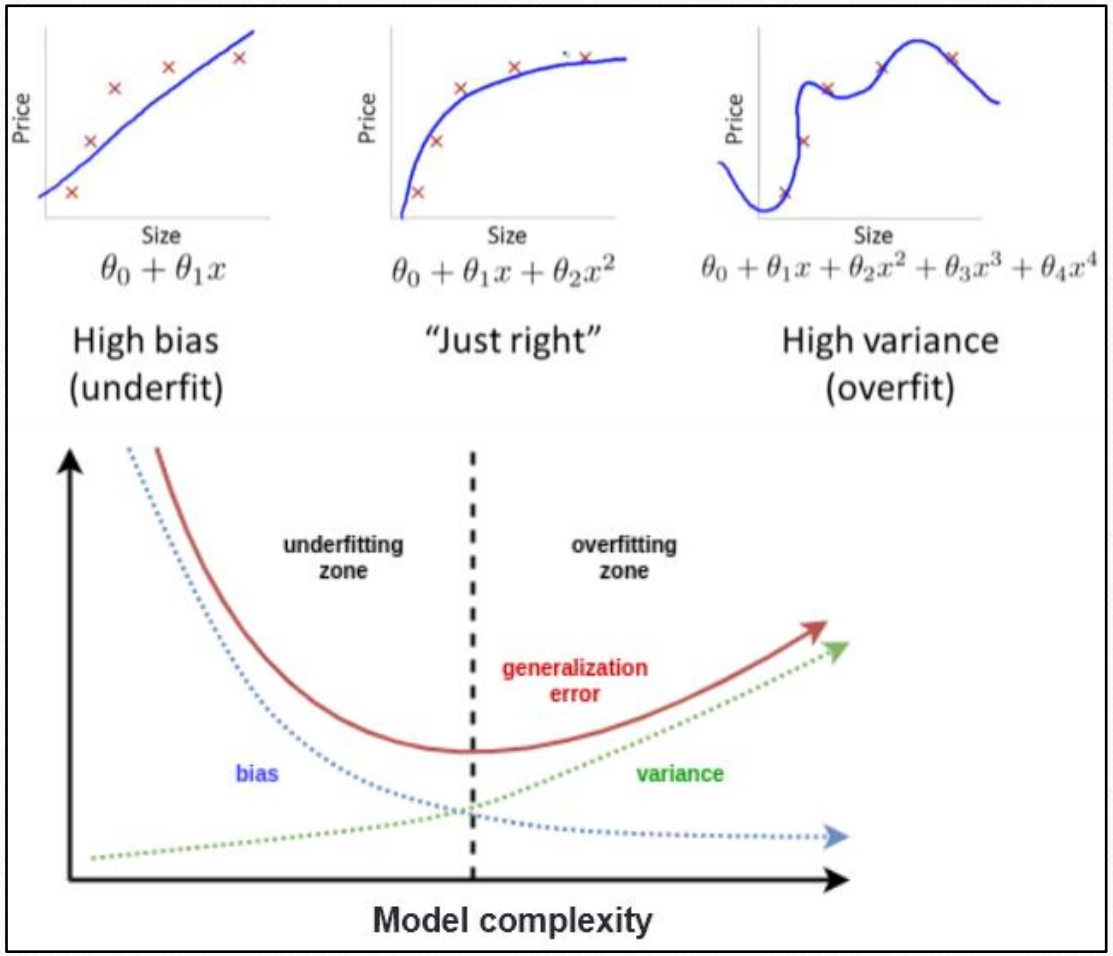


Tradeoff: Bias versus Variance

- There is a tradeoff between a model's ability to minimize *bias* and *variance*.

- Both of these errors depend on the model complexity, e.g. the number of polynomial terms used in a linear regression model.

- As model complexity increases, *bias* will decrease while *variance* will increase. The optimal model complexity can be chosen at the intersection of *bias* and *variance* curves, i.e. with minimum generalization error.



The end